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NARRATIVE-OF-THOUGHT: Improving Temporal Reasoning of Large Language Models via Recounted Narratives

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Abstract

Reasoning about time and temporal relations is an integral aspect of human cognition, essential for perceiving the world and navigating our experiences. Though large language models (LLMs) have demonstrated impressive performance in many reasoning tasks, temporal reasoning remains challenging due to its intrinsic complexity. In this work, we first study an essential task of temporal reasoning—temporal graph generation, to unveil LLMs' inherent, global reasoning capabilities. We show that this task presents great challenges even for the most powerful LLMs, such as GPT-3.5/4. We also notice a significant performance gap by small models ($< 10B$) that lag behind LLMs by 50%. Next, we study how to close this gap with a budget constraint, e.g., not using model finetuning. We propose a new prompting technique tailored for temporal reasoning, NARRATIVE-OF-THOUGHT (NOT), that first converts the events set to a Python class, then prompts a small model to generate a temporally grounded narrative, guiding the final generation of a temporal graph. Extensive experiments showcase the efficacy of NOT in improving various metrics. Notably, NOT attains the highest F1 on the Schema-11 evaluation set, while securing an overall F1 on par with GPT-3.5. NOT also achieves the best structural similarity across the board, even compared with GPT-3.5/4.[1]

1 Introduction

Temporal reasoning is essential for humans to perceive the world, understand daily communications, and interpret the temporal aspects of experiences (Allen, 1983; Nebel and Bürckert, 1995). The recent advent of large language models (LLMs) has garnered substantial attention to their impressive performance in various reasoning tasks, such as arithmetic reasoning (Cobbe et al., 2021; Zhong et al., 2024) and commonsense reasoning (Talmor et al., 2019; Anil et al., 2023). Nonetheless, few LLMs exist to handle temporal reasoning well (Wang and Zhao, 2023; Chu et al., 2023; Chan et al., 2024), due to the task's inherent complexity, mingled with implicit logical inference and the necessity for profound world knowledge.

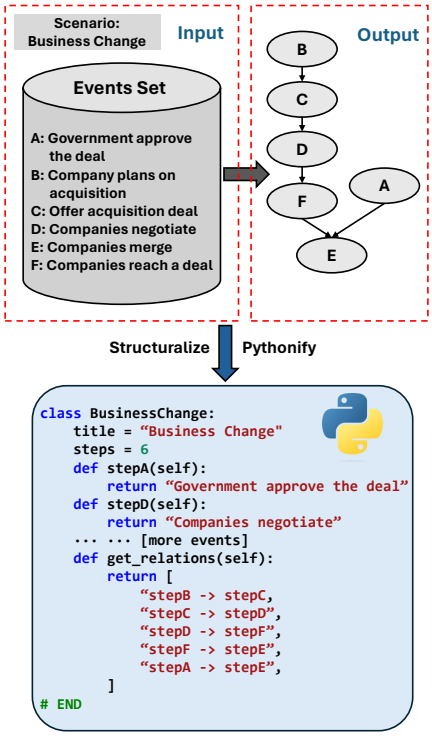

Figure 1: Task overview of **temporal graph generation (TGG)**, where the input is a goal and a set of unordered events. In this work, to better unleash the pre-training power of LLMs trained with a mixture of text and code, we cast TGG as a code completion task.

To gain deeper insights, the research community mainly focuses on two ends along the spectrum: either a simple relation extraction task that orders a pair of events (UzZaman et al., 2013; Yuan et al., 2023), or a perplexing commonsense understanding task demanding multifaceted reasoning skills beyond the mere temporal aspect (Wenzel and Jatowt, 2023; Tan et al., 2023; Xiong et al., 2024).

[1]Our code is available at https://github.com/launchnlp/NoT.

Worse still, the former is limited to a *local* scope spanning two adjacent sentences only and fails to account for the significance of *global* temporal relations, leading to overly optimistic results (Yuan and Liu, 2022; Wang and Zhao, 2023). Therefore, neither setup provides a clear understanding of LLMs' true temporal reasoning abilities.

In this work, we aim to unveil the **inherent, global temporal reasoning capabilities of LLMs**, evaluating them in isolation *free from confounding factors*, and addressing the limitations of previous studies which only focused on local contexts. We first introduce a task of **temporal graph generation (TGG; Figure 1)**: Given a high-level goal[2] $\mathcal{T}$ (e.g., business change) and a set of events $\mathcal{V}$, the objective is to produce a temporal graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$ where a directed edge in $\mathcal{E}$ reveals the temporal order between events. Though this specific notion of TGG is new, many of its applications are not. In this work, we specifically study TGG in order to evaluate and improve the temporal reasoning capability, since TGG is deemed a major bottleneck when LLMs perform temporal reasoning. With TGG, we put forth the first research question.

RQ1: What is the temporal reasoning capability of popular LLMs? Prior work (Wang and Zhao, 2023; Chu et al., 2023) shows a huge gap between AI systems and human performance on various temporal understanding tasks. Additionally, there is a notable performance disparity between proprietary LLMs (e.g., GPT-4) and open-weights LLMs, particularly those with fewer than 10 billion parameters (henceforth, small LLMs). Our study on temporal reasoning reveals a similar trend and identifies the existence of both gaps, as demonstrated in Table 1. This further highlights the importance of an in-depth investigation of TGG, since the performance of downstream tasks (e.g., temporal commonsense understanding) is positively correlated with the inherent, global temporal reasoning capability. Observing the model deficiencies, we are motivated to *fill the gap between open-weights, small LLMs and proprietary large models*. This is due to the fact that open-weights LLMs are generally more accessible, reproducible, and cost-effective to use (Chen et al., 2023; Zhou et al., 2023). In pursuit of this goal, we present the second research question.

RQ2: With a budget constraint (e.g., not allowing further training), how can small LLMs catch up with large models like GPT-3.5/4? Given the constraint that no training will be used, we propose NARRATIVE-OF-THOUGHT (NOT), a special prompting technique tailored for temporal reasoning. This method capitalizes on the recent success of the Chain-of-Thought (CoT) technique (Wei et al., 2022b; Kojima et al., 2022), found effective in solving complex reasoning tasks. To approach TGG, NOT produces a final temporal graph via first generating a *temporally grounded narrative*[3] then sorting the input events topologically in reference to the recounted narrative. Inspired by Madaan et al. (2022); Chen et al. (2022); Gao et al. (2023), NOT also features structural representations by converting the input-output mapping to a Python class, and instructing the generation in code space. We further improve NOT by introducing high-quality reference narratives as part of few-shot demonstrations.

Extensive experiments across three evaluation benchmarks of diverse genres reveal six interesting findings: 1) small LLMs *struggle with temporal reasoning* even with few-shot examples; 2) *CoT is also ineffective at temporal reasoning*, in line with existing finding (Chu et al., 2023); 3) *GPT-4 sometimes falls off the throne due to alignment*, when answering sensitive queries; 4) NOT is a powerful tool to assist small LLMs to catch up with or even *surpass GPT-3.5*, and presents strong compatibility with various base LLMs; 5) *the temporally grounded narratives are significant in improving LLMs' temporal reasoning process*; 6) *AI systems are far from mastering temporal reasoning*, trailing the human baseline by 30 F1 points.

We also analyze the impact of shot numbers and perform a holistic evaluation of reference narratives in few-shot examples. 5-shot is found to be the sweet spot for temporal reasoning, after which the performance plateaus, likely due to long-context challenge. We identify three key characteristics of reference narratives for them to avail small LLMs most: conciseness, simplicity, and factuality.

2 Related Work

2.1 Temporal Reasoning

This work is deeply rooted in a long-standing yet still challenging NLP domain—temporal reasoning (Allen, 1983; Nebel and Bürckert, 1995), which

[2]We use *goal* and *scenario* interchangeably.

[3]In our context, "temporally grounded" refers to events being organized and presented in a way that accurately reflects their temporal sequence or timeline.

involves extraction, representation and reasoning with time and events (Sanampudi and Kumari, 2010). Depending on the cognitive complexity, temporal reasoning in NLP is studied at three levels: temporal expression detection, temporal relation extraction, and temporal graph generation. The simplest **temporal expression detection** task is to identify phrases in the text that convey temporal information (Setzer, 2001; Mani et al., 2001; Pustejovsky et al., 2003), commonly known as TimeX. Further, under-specified TimeX is typically converted to explicit expressions (e.g., "summer 2024") through a process called time expression normalization (Verhagen et al., 2010).

Explicit TimeX is often absent in text, and events usually carry implicit temporal information. To bridge the gap, TempEval (Verhagen et al., 2009; UzZaman et al., 2013) is curated to support the study of **temporal relation extraction**, which aims to detect the temporal relation between two *events* in a document. The most common benchmarks, TB-dense (Chambers et al., 2014) and MATRES (Ning et al., 2018), have witnessed the technique evolution from LSTM (Dligach et al., 2017) and GNN-augmented BERT (Mathur et al., 2021; Wang et al., 2022), to LLMs prompting (Yuan et al., 2023). Yet, these benchmarks are limited by their *locality assumption*, where only pairs of events within a two-sentence window are annotated. Even in this simplified scenario of temporal relation extraction, ChatGPT perform poorly, trailing supervised systems by over 30% (Chan et al., 2024).

The most challenging task, **contextualized temporal graph extraction**, is defined as, given a document, generating a corresponding event-level temporal graph (UzZaman et al., 2013; Madaan and Yang, 2021). This task addresses the limitation of locality by priming models to comprehend the entire article and infer relationships even between distant events. Yet, this area is largely under-investigated, partly due to the scarcity of available datasets. A similar task is **script learning** (Regneri et al., 2010; Modi et al., 2016; Sakaguchi et al., 2021), which targets inducing a stereotypical progression of *complex* events (Schank and Abelson, 1975), represented as a temporal graph of more *atomic* events. This task is usually approached by first extracting information snippets from a given document to build an instance graph, and then expanding the graph to generate a schematic graph using GNN (Li et al., 2021; Jin et al., 2022) or LLM prompting (Dror et al., 2023). Given the remarkable similarities between these two tasks, we instead study a temporal reasoning task formulation that is *fundamental* to both, i.e., **temporal graph generation**. It differs from prior work in at least two dimensions: (1) a limited-context setting, where only abstract event descriptions are available, and (2) only a few training samples at hand, rendering fine-tuning techniques inapplicable. This motivates a *training-free assessment* of LLMs' *inherent, global* temporal reasoning capability.

2.2 Chain-of-Thought and its Variants

Despite the strong problem-solving capability in the general domain (Wei et al., 2022a), LLMs struggle to address more complex reasoning tasks, such as commonsense understanding and arithmetic reasoning (Patel et al., 2021; Talmor et al., 2021a; Huang and Chang, 2023). Wei et al. (2022b) first introduce the concept *Chain-of-Thought (CoT)* by decomposing multi-step problems into intermediate steps. Kojima et al. (2022) further adds a phrase *"Let's think step by step"* to perform zero-shot CoT. These studies underpin the CoT technique in enhancing LLMs' capability for complex reasoning.

Down the line, sophisticated prompting schemes are devised through *structuralization*. One approach is to extend the linear chain structure to Tree-of-Thoughts (Yao et al., 2023) and Graph-of-Thoughts (Besta et al., 2024), enabling expanded exploration space. The huge search space, however, results in a computational resource dilemma. On top of that, leveraging the deterministic execution to narrow the discrepancy between reasoning and final answer, PoT (Chen et al., 2022), PAL (Gao et al., 2023) and Faithful CoT (Lyu et al., 2023) introduce programming languages to describe the reasoning process structurally. These methods are designed exclusively for solving mathematical reasoning and symbolic reasoning, where the reasoning process and computation can be decoupled. In contrast, for temporal reasoning, the reasoning process and the temporal sorting step are intrinsically interleaved. In fact, Chu et al. (2023) has attempted to apply CoT but proved unsuccessful.

Moreover, existing methods are mostly applied to generate intermediate rationales for *simple, atomic outputs*, usually in the format of multi-choice options (Mihaylov et al., 2018; Talmor et al., 2019; Liu et al., 2020), a number (Cobbe et al., 2021; Hendrycks et al., 2021), or yes/no options (Talmor et al., 2021b; Wei et al., 2022a). Our work draws a clear distinction where our focus is